

WEEK 5 OPERATIONAL DIAGNOSTICS REPORT

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PLP Data Analysis with AI

Week 5 Assignment

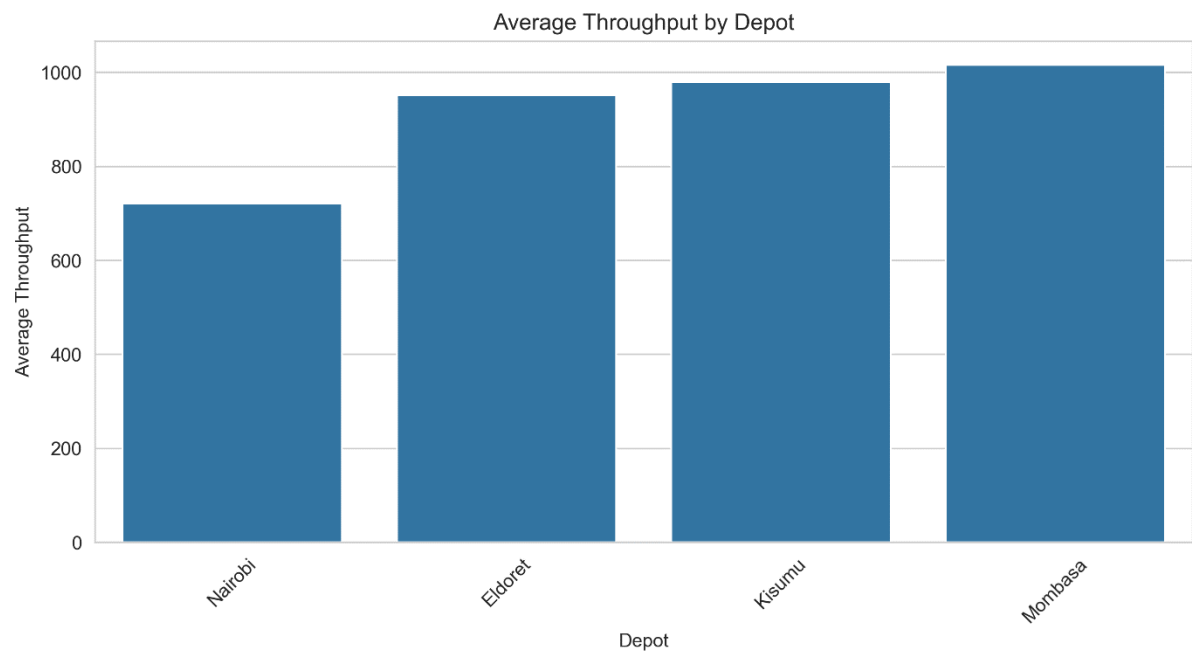
Executive Summary

This report presents the findings of an exploratory data analysis and operational diagnostics investigation conducted on the Mystery Operations dataset. The objective was to identify operational inefficiencies, investigate their underlying causes, and recommend practical actions to improve operational performance. The analysis combined exploratory data analysis, anomaly detection, drill-down analysis, Pareto analysis, and correlation analysis to provide evidence-based operational insights.

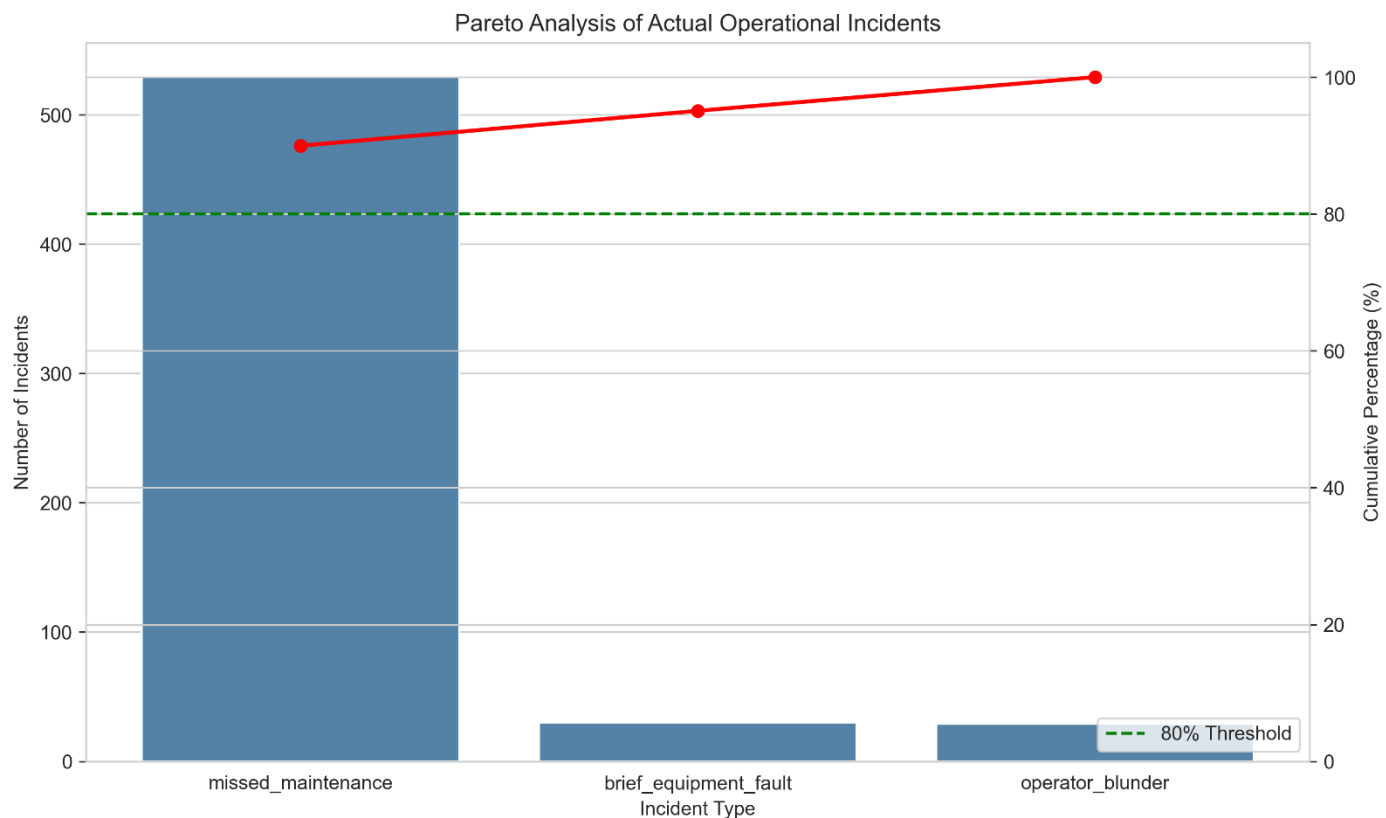
Operational Context

The operational dataset contains production records from multiple depots, including throughput, operating temperature, voltage measurements, maintenance status, work shifts, and incident records. The investigation sought to determine whether production performance differed across depots, identify operational anomalies, and establish the primary factors contributing to operational issues.

KEY FINDINGS

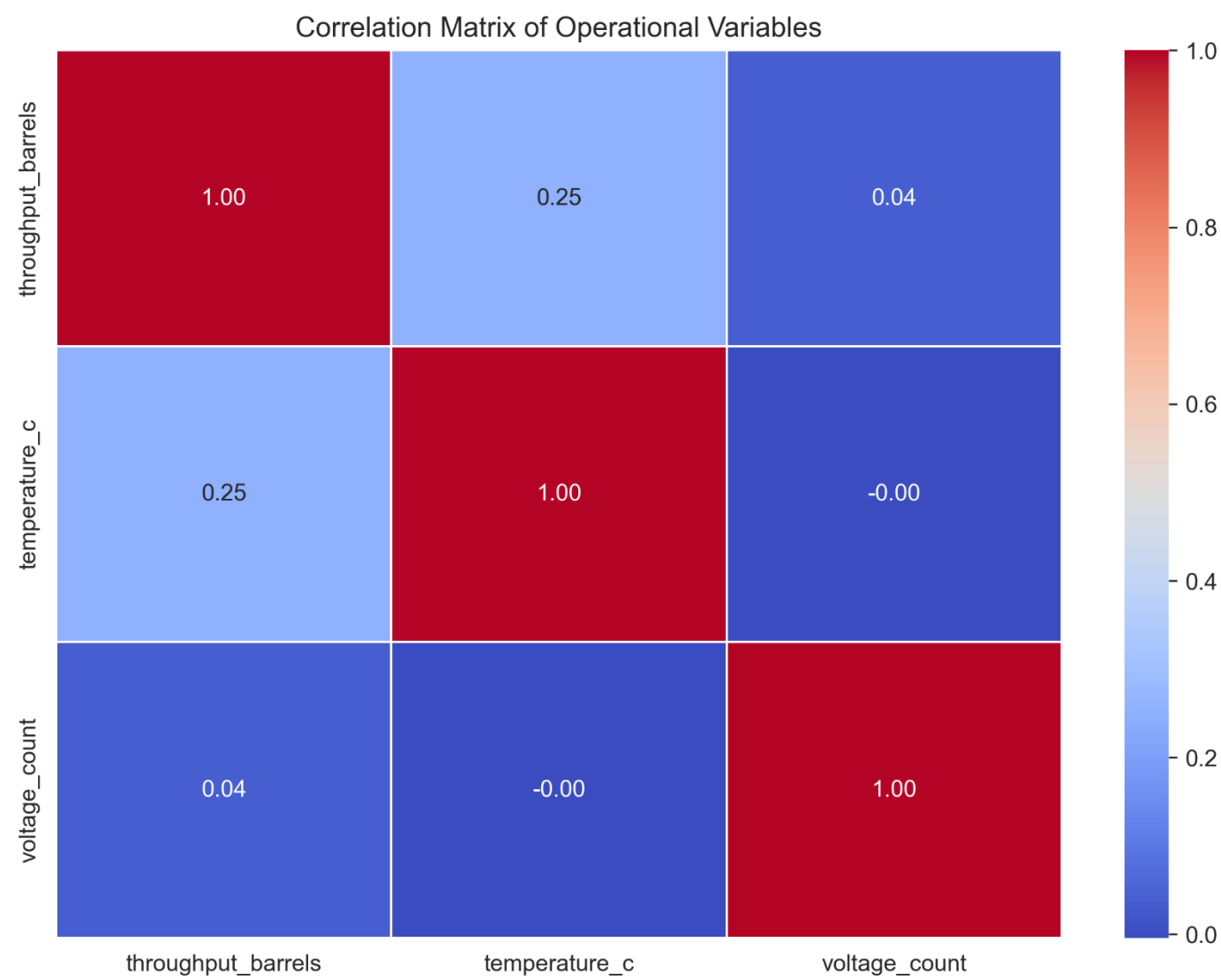


The drill-down analysis revealed considerable variation in production performance across depots. Mombasa recorded the highest average throughput (1015.9 barrels), while Nairobi recorded the lowest average throughput (721.3 barrels). This indicates that operational performance differs significantly between locations and suggests that the Nairobi depot requires further operational review.



Pareto analysis of operational incidents showed that missed maintenance accounts for the majority of recorded incidents, while equipment faults and operator errors occur far less frequently. This finding suggests that improving maintenance planning would likely produce the greatest reduction in operational disruptions.

Correlation Analysis



Correlation analysis found only a weak positive relationship ($r = 0.252$) between operating temperature and production throughput. Voltage showed almost no measurable relationship with throughput ($r = 0.039$). These results indicate that temperature and voltage are unlikely to be the primary drivers of production performance, suggesting that operational practices such as maintenance deserve greater attention.

Root Cause

The investigation indicates that missed maintenance is the primary operational issue affecting performance. Although operational incidents are evenly distributed across depots, Nairobi records substantially lower production throughput than the other depots. This suggests that maintenance effectiveness, operational practices, or resource allocation at the Nairobi depot should be investigated further.

Recommendations

Recommendation 1

Strengthen preventive maintenance scheduling to reduce missed maintenance activities, which represent the largest category of operational incidents.

Recommendation 2

Conduct a detailed operational review of the Nairobi depot to identify the factors contributing to its lower production throughput and compare its operating procedures with those used at the higher-performing Mombasa depot.

Recommendation 3

Continue monitoring production performance through regular operational dashboards that track throughput, maintenance compliance, and incident trends to enable early detection of emerging issues.

Conclusion

The operational diagnostics investigation successfully identified the key operational patterns within the dataset. While temperature and voltage appear to have limited influence on production output, maintenance practices emerge as the most significant operational concern. Addressing missed maintenance and reviewing operational procedures at lower-performing depots will likely result in measurable improvements in production efficiency.